

BT5240 - Computational Systems Biology

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Question 1

1) Assumptions: I have created two separate models, one for minimal media and one for the anaerobic case. The minimal media for the aerobic and anaerobic models are set independently (i.e. They are not the same). Additionally, to set up a model for the anaerobic case, I have considered the lower bound of the O₂ exchange reaction to be 0, thus there can be no “uptake” of O₂ by the system.

a) The biomass fluxes obtained in each of the cases is shown below:

The Biomass Flux in the Aerobic case is: 0.9823718127269904 mmol/gDW/hr

The Biomass Flux in the Anaerobic case is: 0.24150155709717755 mmol/gDW/hr

b) The top 20 reactions with the largest magnitude of fluxes are used for the comparison.

In the Aerobic Case:

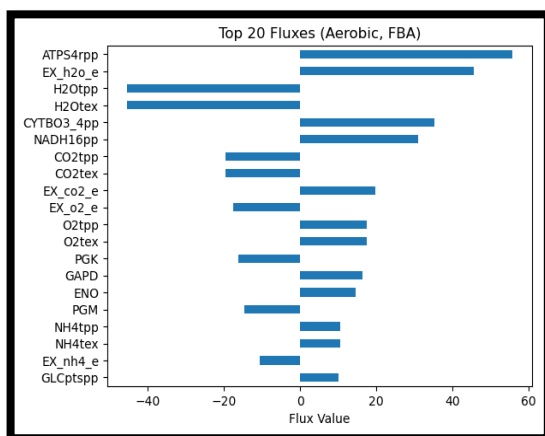


Figure 1.1.b.1

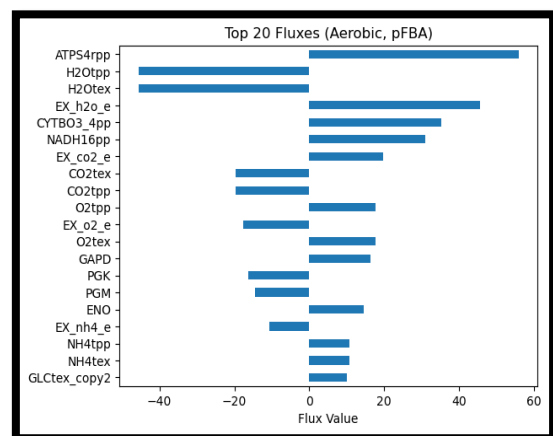


Figure 1.1.b.2

In the Anaerobic Case:

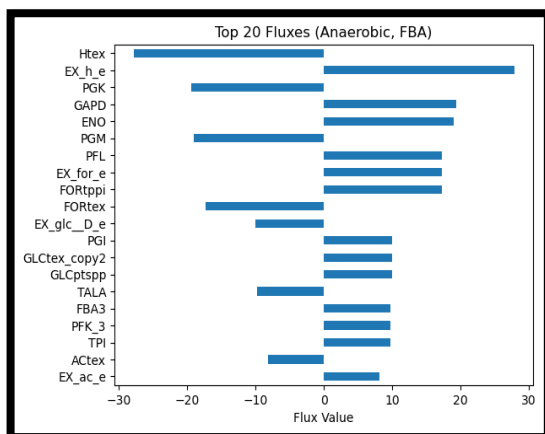


Figure 1.1.b.3

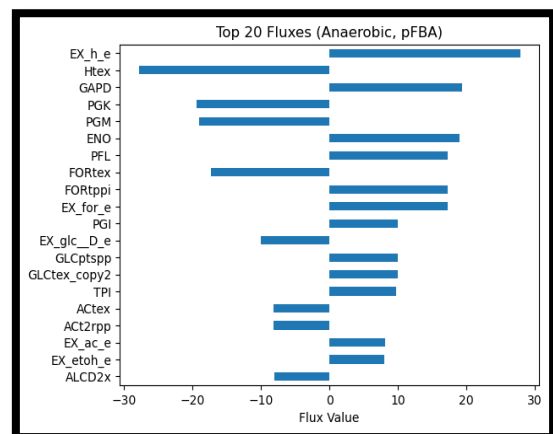


Figure 1.1.b.4

From Figures Figure 1.1.b.1 and 1.1.b.2 and Figures 1.1.b.3 and 1.1.b.4, it is unclear what has changed as all the Fluxes in the top 20 seem to be identical with very minimal changes here and there affecting the order, thus we find the flux difference (FBA-pFBA) and display the top 10 reactions with the largest difference in flux values as shown below:

Reaction_ID	Flux differences (FBA - pFBA)
R1PK	0.916447
R15BPK	0.916447
ACS	0.572306
PTAr	0.572306
SUCDi	0.324964
DHORDfum	0.324964
PAPSR2	0.243502
GTHOr	0.243502
GRXR	0.243502
DMPPS	0.002074

Aerobic Case: Flux Differences

Reaction_ID	Flux differences (FBA - pFBA)
PYK	8.021679
PFK	8.021679
FBA	8.021679
R1PK	0.6340404e
R15BPK	0.6340404
ALATA_L	0.1403984
Htex	5.918821e-12
EX_co2_e	4.170886e-12
PDH	4.101151e-12
FORtex	3.819167e-12

Anaerobic Case: Flux Differences

Thus FBA and pFBA, produce the same Biomass flux, with differences in individual component fluxes!!

c) Reactions that retain wide allowable flux ranges in FVA (in Aerobic and Anaerobic conditions) are shown below. (Note that only the top 50 reactions with the greatest flux range in each condition are shown below)

i) Aerobic conditions; Reactions with a large allowable flux range:

Reaction	Minimum	Maximum	Range
ADK1	-1.000000e+03	1.000000e+03	2000.000000
ADK3	-1.000000e+03	9.992233e+02	1999.223312

Reaction	Minimum	Maximum	Range
NDPK1	-9.992233e+02	1.000000e+03	1999.223312
VPAMTr	-9.995843e+02	9.994289e+02	1999.013190
VALTA	-1.000000e+03	9.990132e+02	1999.013190
ALATA_L	-1.000000e+03	9.990132e+02	1999.013190
THRt2rpp	-1.000000e+03	1.943012e-11	1000.000000
PRPPS	-9.990829e+02	9.171079e-01	1000.000000
ABUTt2pp	0.000000e+00	1.000000e+03	1000.000000
URAt2pp_copy2	-1.000000e+03	0.000000e+00	1000.000000
PPM	-1.000000e+03	4.883321e-13	1000.000000
ASPt2pp_copy2	-1.000000e+03	3.904510e-13	1000.000000
PPAKr	0.000000e+00	1.000000e+03	1000.000000
ICHORS_copy1	-1.000000e+03	6.531265e-14	1000.000000
SERt4pp	0.000000e+00	1.000000e+03	1000.000000
GLDBRAN2	0.000000e+00	1.000000e+03	1000.000000
INDOLEt2pp	0.000000e+00	1.000000e+03	1000.000000
THFAT	0.000000e+00	1.000000e+03	1000.000000
THMDt2pp_copy1	0.000000e+00	1.000000e+03	1000.000000

Reaction	Minimum	Maximum	Range
HPYRI	-1.000000e+03	0.000000e+00	1000.000000
HPYRRx	0.000000e+00	1.000000e+03	1000.000000
GLCtex_copy1	-9.900000e+02	1.000000e+01	1000.000000
ICHORS_copy2	0.000000e+00	1.000000e+03	1000.000000
GLCtex_copy2	0.000000e+00	1.000000e+03	1000.000000
THMDt2pp_copy2	-1.000000e+03	0.000000e+00	1000.000000
CYTDt2pp_copy2	-1.000000e+03	0.000000e+00	1000.000000
ASPt2pp_copy1	0.000000e+00	1.000000e+03	1000.000000
ACt4pp	0.000000e+00	1.000000e+03	1000.000000
ALAt2pp_copy1	0.000000e+00	1.000000e+03	1000.000000
SERt2rpp	-1.000000e+03	0.000000e+00	1000.000000
PTA2	0.000000e+00	1.000000e+03	1000.000000
FOMETRi	0.000000e+00	1.000000e+03	1000.000000
ADNt2pp_copy2	-1.000000e+03	0.000000e+00	1000.000000
GLYCLTt4pp	0.000000e+00	1.000000e+03	1000.000000
GLYt2pp_copy1	0.000000e+00	1.000000e+03	1000.000000
THRt4pp	0.000000e+00	1.000000e+03	1000.000000

Reaction	Minimum	Maximum	Range
CYTDt2pp_copy1	0.000000e+00	1.000000e+03	1000.000000
INDOLEt2rpp	-3.865381e-14	1.000000e+03	1000.000000
ADNt2pp_copy1	0.000000e+00	1.000000e+03	1000.000000
GLBRAN2	0.000000e+00	1.000000e+03	1000.000000
GLYt2pp_copy2	-1.000000e+03	0.000000e+00	1000.000000
ALAt2pp_copy2	-1.000000e+03	0.000000e+00	1000.000000
GLYCLTt2rpp	-1.000000e+03	0.000000e+00	1000.000000
GLYt4pp	0.000000e+00	1.000000e+03	1000.000000
URAt2pp_copy1	0.000000e+00	1.000000e+03	1000.000000
ALAt4pp	0.000000e+00	1.000000e+03	1000.000000
INSt2pp_copy2	-1.000000e+03	0.000000e+00	1000.000000
URIt2pp_copy1	0.000000e+00	1.000000e+03	1000.000000
NAt3pp	0.000000e+00	1.000000e+03	1000.000000
INSt2pp_copy1	0.000000e+00	1.000000e+03	1000.000000

ii) Anaerobic conditions; Reactions with a large allowable flux range:

Reaction	Minimum	Maximum	Range
ADK1	-1.000000e+03	1.000000e+03	2000.000000

Reaction	Minimum	Maximum	Range
ADK3	-1.000000e+03	9.998091e+02	1999.809063
NDPK1	-9.998091e+02	1.000000e+03	1999.809063
VALTA	-1.000000e+03	9.997574e+02	1999.757407
VPAMTr	-9.998978e+02	9.998596e+02	1999.757407
ALATA_L	-1.000000e+03	9.997574e+02	1999.757407
PRPPS	-9.997744e+02	2.255651e-01	1000.000000
CRNDt2rpp	-1.622170e-12	1.000000e+03	1000.000000
ASPt2pp_copy2	-1.000000e+03	8.153766e-13	1000.000000
GLCtex_copy1	-9.900000e+02	1.000000e+01	1000.000000
CRNt2rpp	-1.000000e+03	0.000000e+00	1000.000000
TRSARr	-1.000000e+03	4.995632e-13	1000.000000
INDOLEt2rpp	-3.209117e-13	1.000000e+03	1000.000000
URIt2pp_copy2	-1.000000e+03	0.000000e+00	1000.000000
SERt2rpp	-1.000000e+03	0.000000e+00	1000.000000
URAt2pp_copy2	-1.000000e+03	0.000000e+00	1000.000000
ICHORS_copy1	-1.000000e+03	1.406038e-13	1000.000000
INDOLEt2pp	0.000000e+00	1.000000e+03	1000.000000

Reaction	Minimum	Maximum	Range
CYTDT2pp_copy2	-1.000000e+03	0.000000e+00	1000.000000
GLYt4pp	0.000000e+00	1.000000e+03	1000.000000
THMDt2pp_copy2	-1.000000e+03	0.000000e+00	1000.000000
GLBRAN2	0.000000e+00	1.000000e+03	1000.000000
R1PK	0.000000e+00	1.000000e+03	1000.000000
GLDBRAN2	0.000000e+00	1.000000e+03	1000.000000
GLUABUTt7pp	0.000000e+00	1.000000e+03	1000.000000
GLUt4pp	0.000000e+00	1.000000e+03	1000.000000
PROt2rpp	-1.000000e+03	9.699272e-17	1000.000000
GLCtex_copy2	0.000000e+00	1.000000e+03	1000.000000
ABUTt2pp	0.000000e+00	1.000000e+03	1000.000000
GLYt2pp_copy1	0.000000e+00	1.000000e+03	1000.000000
GLYt2pp_copy2	-1.000000e+03	0.000000e+00	1000.000000
HPYRI	-1.000000e+03	0.000000e+00	1000.000000
FOMETRi	0.000000e+00	1.000000e+03	1000.000000
HPYRRx	0.000000e+00	1.000000e+03	1000.000000
ICHORS_copy2	0.000000e+00	1.000000e+03	1000.000000

Reaction	Minimum	Maximum	Range
INSt2pp_copy1	0.000000e+00	1.000000e+03	1000.000000
SERt4pp	0.000000e+00	1.000000e+03	1000.000000
INSt2pp_copy2	-1.000000e+03	0.000000e+00	1000.000000
PTA2	0.000000e+00	1.000000e+03	1000.000000
NAt3pp	0.000000e+00	1.000000e+03	1000.000000
PPAKr	0.000000e+00	1.000000e+03	1000.000000
PPCSCT	0.000000e+00	1.000000e+03	1000.000000
THRt4pp	0.000000e+00	1.000000e+03	1000.000000
PROt4pp	0.000000e+00	1.000000e+03	1000.000000
URIt2pp_copy1	0.000000e+00	1.000000e+03	1000.000000
CYTDt2pp_copy1	0.000000e+00	1.000000e+03	1000.000000
ALAt2pp_copy1	0.000000e+00	1.000000e+03	1000.000000
ALAt2pp_copy2	-1.000000e+03	0.000000e+00	1000.000000
ALAt4pp	0.000000e+00	1.000000e+03	1000.000000
URAt2pp_copy1	0.000000e+00	1.000000e+03	1000.000000

d) The distinct insights provided by FBA, pFBA, and FVA are listed below:

FBA: Typically used to predict or estimate the theoretical (stoichiometric) upper bound of the production of a particular metabolite of interest by the cell.

As an example, say a pharmaceutical company wishes to switch to a new cell-line for the production of Camptothecin. They can set the objective function as Camptothecin in the Genome-Scale Metabolic model of this cell and infer whether the new cell-line can even meet the requirements to be considered commercially feasible. The results might not be accurate (as the cell may prioritize biomass production, energy conservation etc.), but they give us a good idea on upper limits to make decisions.

pFBA: Minimizes the sparsity of the flux vector while ensuring optimal biomass production. This is interpreted as the cell maximizing its growth using the least amount of resources possible (which in some cases can capture the system very accurately).

This exact minimization of the L1 norm can be clearly seen in Q1, 1, b), where the flux values in pFBA are lesser compared to FBA, particularly in the anaerobic case.

Additionally, pFBA also ensures that all unnecessary cycles/loops in the reaction space have no flux through them (that is they become zero).

FVA: From FBA, we are able to gather what is the theoretical maximum scope of productivity of a given cell-line, using FVA we are able to determine what is a more realistic maximum scope of productivity for the same cell-line.

Since maximizing biomass is considered a good objective function (that logically speaking most cells would tend to do). Now, we treat this “maximum biomass” as one additional constrain and try to explore the more constrained solution space in the flux space. A high productivity with very minimal range of flux for our product is preferred as it implies we can expect high-yield with higher confidence. Thus, FVA gives us an estimate of how likely the outcome we require occurs.

2) Assumptions: Minimal media is not assumed for this section of the question.

a) Three weight ratios were used for the comparison, and the objective function is given below:

I) {biomass: 0.97, product: 0.03}

II) {biomass: 0.95, product: 0.05}

III) {biomass: 0.935, product: 0.065}

In the Aerobic Case:

The weights are 0.97 for biomass and 0.03 for the product

The value of the new objective function is: 0.9529006583459453

Biomass reaction flux: 0.9823718127277786

Ethanol reaction flux: 0.0

The weights are 0.95 for biomass and 0.05 for the product

The value of the new objective function is: 1.0437348139197116

Biomass reaction flux: 0.1922759514329774

Ethanol reaction flux: 17.22145320116766

The weights are 0.935 for biomass and 0.065 for the product

The value of the new objective function is: 1.2999999999999996

Biomass reaction flux: -6.836723420872428e-17

Ethanol reaction flux: 19.999999999999993

In the Anaerobic Case:

The weights are 0.97 for biomass and 0.03 for the product

The value of the new objective function is: 0.7001772605247347

Biomass reaction flux: 0.190262338779783

Ethanol reaction flux: 17.187426396944844

The weights are 0.95 for biomass and 0.05 for the product

The value of the new objective function is: 1.040120541688038

Biomass reaction flux: 0.19026233877978405

Ethanol reaction flux: 17.18742639694486

The weights are 0.935 for biomass and 0.065 for the product

The value of the new objective function is: 1.30000000000000074

Biomass reaction flux: 1.4229223955576858e-15

Ethanol reaction flux: 20.0000000000000092

b) To analyse the change in the Overall Flux distribution, the top 20 reactions with the largest magnitude of fluxes in each case are used for the comparison.

In the Aerobic Case:

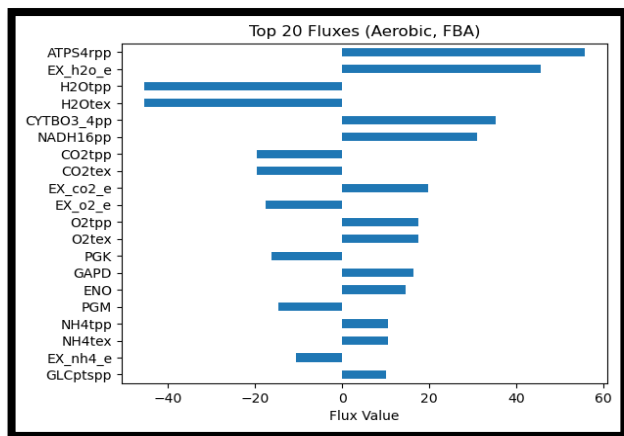


Figure 1.2.b.1: Maximize Biomass

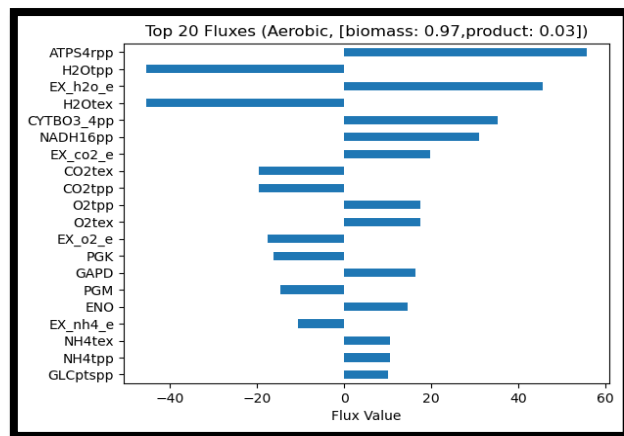


Figure 1.2.b.2: Modified Objective 1

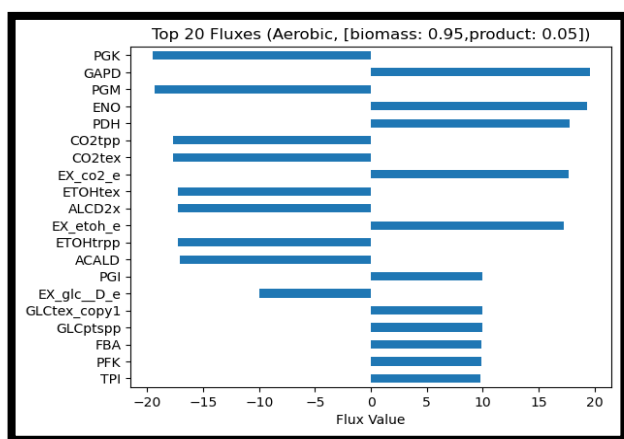


Figure 1.2.b.3: Modified Objective 2

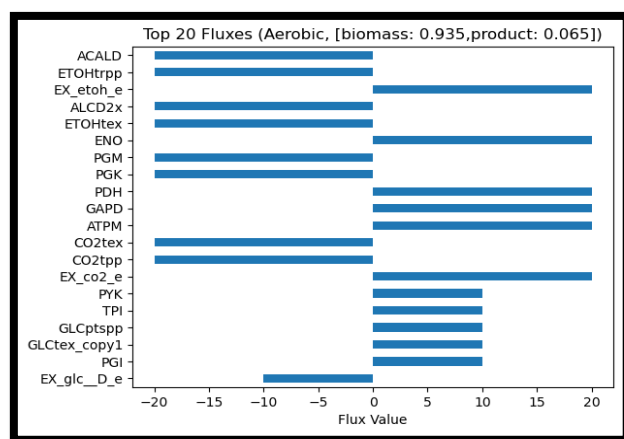


Figure 1.2.b.4: Modified Objective 3

The biomass flux rapidly falls for miniscule changes in the ratio in the objective function. Conversely, ethanol flux rapidly increases with minute changes in the objective function.

Interestingly, with increasing ethanol production the cells CO₂ efflux also increases, however its O₂ and Glucose uptake remain fairly stable.

In the Anaerobic Case:

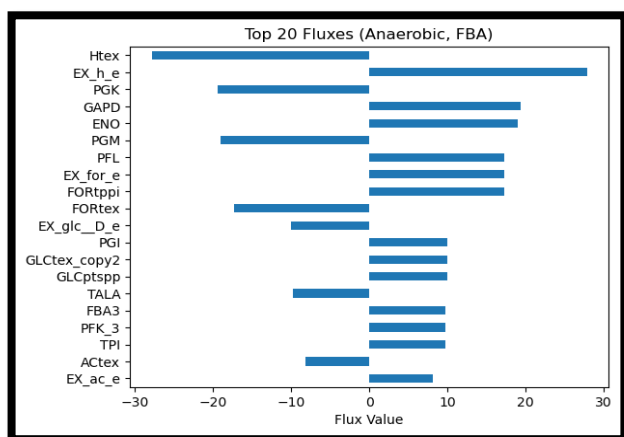


Figure 1.2.b.5: Maximize Biomass

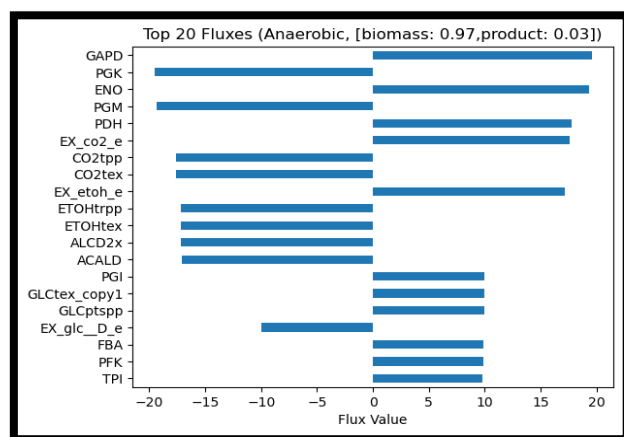


Figure 1.2.b.6: Modified Objective 1

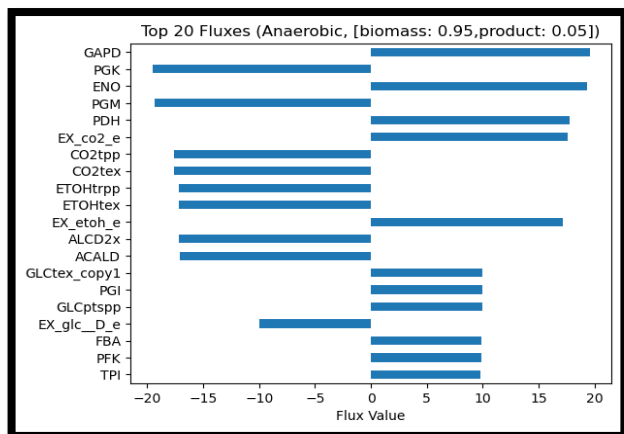


Figure 1.2.b.7: Modified Objective 2

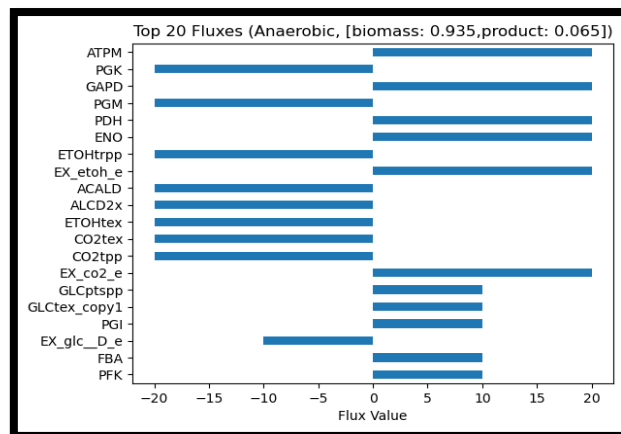


Figure 1.2.b.8: Modified Objective 3

The biomass flux changes quite gradually with changes to the objective function when compared to the aerobic case. And a similar case is seen with ethanol flux as well.

Electron exchange reactions appear to dominate in the absence of O_2 as an electron acceptor.

In general reactions involved in the glycolysis pathway such as PGK, PFK, and PYK have an increased flux in both (aerobic and anaerobic) cases when ethanol production is increased.

c) In both cases we see an increase in flux in the glycolysis pathway and fermentation pathway (which is logical considering ethanol is a fermentation product).

However, we also see that PDH (Pyruvate dehydrogenase) also carrying a high flux through it in the aerobic case, indicating that a good amount of glucose is being directed into the TCA cycle and OXPHOS pathways.

Though the maximum amount of EtOH produced in both cases appears to be the same at nearly 20 $\text{mmol} \cdot \text{gDW}^{-1} \cdot \text{h}^{-1}$, we see in the aerobic case that the cell is not using the entire carbon flux for EtOH production.

Question 2

1) Assumptions: To compare lifestyle of the two organisms, references [1, 2] were used.

a) The number of genes, metabolites, and reactions in each organism as well as its reaction-to-gene ratio was computed and is shown below:

The number of genes in the Mycobacterium tuberculosis: 661

The number of reactions in the Mycobacterium tuberculosis: 1025

The number of metabolites in the Mycobacterium tuberculosis: 825

The reaction-to-gene ratio for Mycobacterium tuberculosis is: 1.5506807866868382

The number of genes in the *Acinetobacter baumannii*: 709

The number of reactions in the *Acinetobacter baumannii*: 1015

The number of metabolites in the *Acinetobacter baumannii*: 888

The reaction-to-gene ratio for *Acinetobacter baumannii* is: 1.4315937940761636

It is evident that a higher number of genes doesn't correspond to a higher number of reactions.

Acinetobacter baumannii has a greater number of genes, however its reaction-to-gene ratio is lesser than that of *Mycobacterium tuberculosis* which has nearly 50 genes lesser.

However, a key inference that can potentially be made is that the total number of reactions regardless of the organism, roughly remains the same (assuming these are phylogenetically close that is).

b) *Mycobacterium tuberculosis* is a species of pathogenic Gram-positive bacteria, responsible for causing tuberculosis. *Acinetobacter baumannii* is defined as a non-motile, Gram-negative coccobacillus that acts as an opportunistic pathogen, primarily affecting hospitalized patients and those with compromised immune system.

Mycobacterium tuberculosis has a much slower growth rate, develops an antibiotic resistance at a moderate rate, and can survive in various types of environments. Contrasting this to *Acinetobacter baumannii* which has a moderate growth rate, with the ability to develop antibiotic resistance exceptionally quickly, and survives only in hospitals.

As such the key difference we expect to see in the above would be based on the metabolites used by each organism. Additionally, we should expect to see reaction pathway/flux differences as well, given the fact that the metabolism of these organisms is quite dissimilar. The main differences in the fluxes should arise from: Biomass flux, Cell Wall Synthesis/Maintenance reactions, and Anabolic reactions.

2) Assumptions: To determine whether a gene is essential or not, an arbitrary threshold for growth rate had to be considered. This was assumed to be 5% of the maximal growth rate possible for the organism.

a) The essential genes for both organisms were determined and the output is shown below:

i) *Mycobacterium tuberculosis*

{'Rv1285', 'Rv2392', 'Rv0391', 'Rv2122c', 'Rv1202', 'Rv1603', 'Rv0382c', 'Rv1612', 'Rv3815c', 'Rv2384', 'Rv1601', 'Rv1878', 'Rv1385', 'Rv1981c', 'Rv2335', 'Rv3809c', 'Rv1449c', 'Rv2210c', 'Rv3808c', 'Rv1302', 'Rv1656', 'Rv3708c', 'Rv2935', 'Rv0618', 'Rv2928', 'Rv1512', 'Rv1655', 'Rv2383c', 'Rv1606', 'Rv2163c', 'Rv2201', 'Rv1654', 'Rv3393', 'Rv3265c', 'Rv3340', 'Rv2957', 'Rv3423c', 'Rv1201c', 'Rv2763c', 'Rv1484', 'Rv0334', 'Rv3002c', 'Rv0189c', 'Rv0993', 'Rv0809', 'Rv2386c', 'Rv2987c', 'Rv3227', 'Rv1843c', 'Rv2932',

'Rv3490', 'Rv1551', 'Rv2155c', 'Rv3913', 'Rv1381', 'Rv1662', 'Rv0644c', 'Rv2178c', 'Rv2940c', 'Rv1323', 'Rv2465c', 'Rv1018c', 'Rv1392', 'Rv1079', 'Rv1653', 'Rv0357c', 'Rv2860c', 'Rv3581c', 'Rv2381c', 'Rv1294', 'Rv2157c', 'Rv1611', 'Rv1493', 'Rv3795', 'Rv0957', 'Rv3582c', 'Rv2051c', 'Rv2933', 'Rv2153c', 'Rv2612c', 'Rv0777', 'Rv2136c', 'Rv2958c', 'Rv0733', 'Rv0470c', 'Rv2158c', 'Rv0620', 'Rv2503c', 'Rv3280', 'Rv0803', 'Rv2445c', 'Rv3410c', 'Rv1086', 'Rv2220', 'Rv1605', 'Rv2222c', 'Rv1350', 'Rv3838c', 'Rv2156c', 'Rv2378c', 'Rv3372', 'Rv1885c', 'Rv2524c', 'Rv1613', 'Rv2245', 'Rv1293', 'Rv3465', 'Rv2941', 'Rv0619', 'Rv2539c', 'Rv3793', 'Rv3709c', 'Rv1011', 'Rv2246', 'Rv0437c', 'Rv2538c', 'Rv2124c', 'Rv3754', 'Rv0436c', 'Rv1492', 'Rv1652', 'Rv2552c', 'Rv1663', 'Rv2380c', 'Rv2192c', 'Rv2988c', 'Rv2152c', 'Rv3628', 'Rv2949c', 'Rv2931', 'Rv3266c', 'Rv2947c', 'Rv0780', 'Rv0788', 'Rv1017c', 'Rv3710', 'Rv0884c', 'Rv2726c', 'Rv0482', 'Rv3255c', 'Rv2753c', 'Rv0858c', 'Rv2006', 'Rv1338', 'Rv2610c', 'Rv2611c', 'Rv3248c', 'Rv3068c', 'Rv2540c', 'Rv2934', 'Rv1383', 'Rv2995c', 'Rv2139', 'Rv3396c', 'Rv1692', 'Rv0772', 'Rv3048c', 'Rv0046c', 'Rv3051c', 'Rv2537c', 'Rv2962c', 'Rv3800c', 'Rv1699', 'Rv2870c', 'Rv2773c', 'Rv0500', 'Rv2981c', 'Rv3794', 'Rv1286', 'Rv3792', 'Rv3818', 'Rv2379c', 'Rv3001c', 'Rv2382c', 'Rv2121c', 'Rv2881c', 'Rv1315', 'Rv1380', 'Rv1599', 'Rv1389', 'Rv3264c', 'Rv3801c', 'Rv3436c', 'Rv2504c', 'Rv3398c', 'Rv0570', 'Rv3411c', 'Rv0855'}

ii) *Acinetobacter baumannii*

{'ABAYE_RS02500', 'ABAYE_RS18965', 'ABAYE_RS02015', 'ABAYE_RS14110', 'ABAYE_RS12520', 'ABAYE_RS12850', 'ABAYE_RS06470', 'ABAYE_RS04905', 'ABAYE_RS02010', 'ABAYE_RS08240', 'ABAYE_RS01360', 'ABAYE_RS17020', 'ABAYE_RS00995', 'ABAYE_RS03025', 'ABAYE_RS15285', 'ABAYE_RS17645', 'ABAYE_RS15425', 'ABAYE_RS00990', 'ABAYE_RS01015', 'ABAYE_RS10010', 'ABAYE_RS00740', 'ABAYE_RS04830', 'ABAYE_RS07485', 'ABAYE_RS15605', 'ABAYE_RS02990', 'ABAYE_RS08665', 'ABAYE_RS13000', 'ABAYE_RS01365', 'ABAYE_RS18870', 'ABAYE_RS11080', 'ABAYE_RS02405', 'ABAYE_RS06075', 'ABAYE_RS07990', 'ABAYE_RS01955', 'ABAYE_RS04585', 'ABAYE_RS14875', 'ABAYE_RS15510', 'ABAYE_RS18920', 'ABAYE_RS16260', 'ABAYE_RS15295', 'ABAYE_RS15465', 'ABAYE_RS17250', 'ABAYE_RS04130', 'ABAYE_RS17635', 'ABAYE_RS05235', 'ABAYE_RS06005', 'ABAYE_RS05165', 'ABAYE_RS17215', 'ABAYE_RS06170', 'ABAYE_RS04795', 'ABAYE_RS05960', 'ABAYE_RS01060', 'ABAYE_RS01280', 'ABAYE_RS12515', 'ABAYE_RS06300', 'ABAYE_RS02880', 'ABAYE_RS01285', 'ABAYE_RS01960', 'ABAYE_RS19090', 'ABAYE_RS15585', 'ABAYE_RS18860', 'ABAYE_RS10300', 'ABAYE_RS02090', 'ABAYE_RS05125', 'ABAYE_RS03620', 'ABAYE_RS06435', 'ABAYE_RS08555', 'ABAYE_RS10535', 'ABAYE_RS08660', 'ABAYE_RS01965', 'ABAYE_RS17535', 'ABAYE_RS03915', 'ABAYE_RS14710', 'ABAYE_RS02420'}

b)

Since gene annotations (using gene.annotations) didn't contain similar fields for both the models for the vast majority of genes, we have decided to use the genes belonging to common reactions between the 2 models to determine the essential genes ("So, you can use reactions in which the genes participate as a proxy to identify shared genes." – TAs of BT5240). The output is shown below:

Reaction Name Mycobacterium tuberculosis**Acinetobacter baumannii**

0	DGK1	[Rv1389]	[ABAYE_RS02090]
1	NDPK9	[Rv2445c]	[ABAYE_RS16260]
2	DAPE	[Rv2726c]	[ABAYE_RS04830]
3	RNDR3	[Rv3051c, Rv3048c]	[ABAYE_RS15295, ABAYE_RS15285]
4	DASYN160	[Rv2881c]	[ABAYE_RS08240]
5	NDPK2	[Rv2445c]	[ABAYE_RS16260]
6	CTPS1	[Rv1699]	[ABAYE_RS08660]
7	PSSA160	[Rv0436c]	[ABAYE_RS02880]
8	NDPK7	[Rv2445c]	[ABAYE_RS16260]
9	GALUi	[Rv0993]	[ABAYE_RS18870]
10	DHQS	[Rv2538c]	[ABAYE_RS02015]
11	UAAGDS	[Rv2158c]	[ABAYE_RS01955]
12	NDPK5	[Rv2445c]	[ABAYE_RS16260]
13	PSERT	[Rv0884c]	[ABAYE_RS04905]
14	DHDPRy	[Rv2773c]	[ABAYE_RS00740]
15	PAPPT3	[Rv2156c]	[ABAYE_RS01965]
16	UDCPDP	[Rv2136c]	[ABAYE_RS04130]

Reaction Name Mycobacterium tuberculosis**Acinetobacter baumannii**

17	THDPS	[Rv1201c]	[ABAYE_RS05125]
18	UGMDDS	[Rv2157c]	[ABAYE_RS01960]
19	RNDR1	[Rv0570, Rv1981c]	[ABAYE_RS15295, ABAYE_RS15285]
20	NDPK3	[Rv2445c]	[ABAYE_RS16260]
21	NDPK4	[Rv2445c]	[ABAYE_RS16260]
22	G1PACT	[Rv1018c]	[ABAYE_RS00995]
23	OHPBAT	[Rv0884c]	[ABAYE_RS04905]
24	SDPDS	[Rv1202]	[ABAYE_RS03915]
25	NDPK6	[Rv2445c]	[ABAYE_RS16260]
26	GF6PTA	[Rv3436c]	[ABAYE_RS00990]
27	PPND	[Rv3754]	[ABAYE_RS06470]
28	SHKK	[Rv2539c]	[ABAYE_RS02010]
29	SDPTA	[Rv0858c]	[ABAYE_RS11080]
30	UAGCVT	[Rv1315]	[ABAYE_RS15605]
31	PSD160	[Rv0437c]	[ABAYE_RS01060]
32	UAMAGS	[Rv2155c]	[ABAYE_RS17535]
33	NDPK1	[Rv2445c]	[ABAYE_RS16260]

Reaction Name	<i>Mycobacterium tuberculosis</i>	<i>Acinetobacter baumannii</i>
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34	UAPGR	[Rv0482]	[ABAYE_RS07990]
35	NDPK8	[Rv2445c]	[ABAYE_RS16260]
36	SHK3Dr	[Rv2552c]	[ABAYE_RS02405]
37	RNDR2	[Rv3051c, Rv3048c]	[ABAYE_RS15295, ABAYE_RS15285]
38	ALAALAr	[Rv2981c]	[ABAYE_RS01285]
39	UAGDP	[Rv1018c]	[ABAYE_RS00995]
40	PSCVT	[Rv3227]	[ABAYE_RS06470]
41	UAMAS	[Rv2152c]	[ABAYE_RS01280]
42	RNDR4	[Rv0570, Rv1981c]	[ABAYE_RS15295, ABAYE_RS15285]
43	CHORS	[Rv2540c]	[ABAYE_RS10010]
44	CTPS2	[Rv1699]	[ABAYE_RS08660]

As shown in the code, this model doesn't have annotations (or subsystems), as such I manually identify the pathways the first 20 reactions belong to, as shown below:

S.No	Reaction ID	Reaction	Classification
1	DGK1	$\text{atp_c} + \text{dgmp_c} \rightleftharpoons \text{adp_c} + \text{dgdp_c}$	Nucleotide Interconversion
2	NDPK9	$\text{atp_c} + \text{idp_c} \rightleftharpoons \text{adp_c} + \text{itp_c}$	Nucleotide Metabolism
3	DAPE	$26\text{dap_LL_c} \rightleftharpoons 26\text{dap_M_c}$	Thr and Lys Metabolism
4	RNDR3	$\text{cdp_c} + \text{trdrd_c} \rightarrow \text{dcdp_c} + \text{h2o_c} + \text{trdox_c}$	Nucleotide Salvage Pathway
5	DASYN160	$\text{ctp_c} + \text{h_c} + \text{pa160_c} \rightleftharpoons \text{cdpdhdecg_c} + \text{ppi_c}$	Glycerophospholipid Metabolism
6	NDPK2	$\text{atp_c} + \text{udp_c} \rightleftharpoons \text{adp_c} + \text{utp_c}$	Pyrimidine metabolism
7	CTPS1	$\text{atp_c} + \text{nh4_c} + \text{utp_c} \rightleftharpoons \text{adp_c} + \text{ctp_c} + 2.0 \text{ h_c} + \text{pi_c}$	Nucleotide Metabolism

8	PSSA160	$\text{cdpdhdecg_c} + \text{ser_L_c} \rightleftharpoons \text{cmp_c} + \text{h_c} + \text{ps160_c}$	Glycerophospholipid Metabolism
9	NDPK7	$\text{atp_c} + \text{dcdp_c} \rightleftharpoons \text{adp_c} + \text{dctp_c}$	Pyrimidine Metabolism
10	GALUi	$\text{glp_c} + \text{h_c} + \text{utp_c} \rightleftharpoons \text{ppi_c} + \text{udpg_c}$	Galactose metabolism
11	DHQS	$2\text{dda7p_c} \rightarrow 3\text{dhq_c} + \text{pi_c}$	Tyr, Trp, and Phe Metabolism
12	UAAGDS	$26\text{dap_M_c} + \text{atp_c} + \text{uamag_c} \rightleftharpoons \text{adp_c} + \text{h_c} + \text{pi_c} + \text{ugmd_c}$	Cell Envelope Biosynthesis
13	NDPK5	$\text{atp_c} + \text{dgdp_c} \rightleftharpoons \text{adp_c} + \text{dgtp_c}$	Nucleotide Salvage Pathway
14	PSERT	$3\text{php_c} + \text{glu_L_c} \rightleftharpoons \text{akg_c} + \text{pser_L_c}$	Gly and Ser Metabolism
15	DHDPRy	$23\text{dhdp_c} + \text{h_c} + \text{nadph_c} \rightleftharpoons \text{nadp_c} + \text{thdp_c}$	Thr and Lys Metabolism
16	PAPPT3	$\text{udcpp_c} + \text{ugmda_c} \rightleftharpoons \text{uagmda_c} + \text{ump_c}$	Cell Envelope Biosynthesis
17	UDCPDP	$\text{h2o_c} + \text{udcpdp_c} \rightleftharpoons \text{h_c} + \text{pi_c} + \text{udcpp_c}$	Cell Envelope Biosynthesis
18	THDPS	$\text{h2o_c} + \text{succoa_c} + \text{thdp_c} \rightleftharpoons \text{coa_c} + \text{sl2a6o_c}$	Lys biosynthesis
19	UGMDDS	$\text{alaala_c} + \text{atp_c} + \text{ugmd_c} \rightleftharpoons \text{adp_c} + \text{h_c} + \text{pi_c} + \text{ugmda_c}$	Cell Envelope Biosynthesis
20	RNDR1	$\text{adp_c} + \text{trdrd_c} \rightarrow \text{dadp_c} + \text{h2o_c} + \text{trdox_c}$	Nucleotide interconversion

NOTE: In the above table reactions are obtained from BiGG Models Universal Reaction pages and reaction's classifications were obtained from BiGG Models of other organisms (since our models do not have any subsystems associated with them) with the same reaction.

3) From the above table we see that the DHQS reaction is linked with Amino-Acid Metabolism, as such we shall choose to analyse the genes associated with this reaction in the following sections of this question (Rv2358c, ABAYE_RS02015).

a) We find that the gene is **3-dehydroquinate synthase** [3]. 3-Dehydroquinate synthase's function is to catalyse the conversion of 3-deoxy-D-arabino-heptulosonate 7-phosphate into 3-dehydroquinate by utilizing NAD⁺ and a cobalt cofactor [4].

This enzyme catalyses the second key step in the shikimate pathway, which is responsible for the biosynthesis of aromatic amino acids (tryptophan, phenylalanine, and tyrosine). Inhibition of this enzyme therefore inhibits the synthesis of aromatic amino acids, which are essential for protein synthesis, in bacteria, archaea, fungi, algae, some protozoans which ultimately leads to their (*Mycobacterium tuberculosis* and *Acinetobacter baumannii*) deaths. Crucially, this enzyme has the makings to be a potential target for antimicrobial agents as this pathway is not found in mammals [5].

b) Multidrug Resistance (MDR) in the case of both *Mycobacterium tuberculosis* and *Acinetobacter baumannii* is on a rampant increase, and treating MDR strains requires a longer and more complex treatment regimens, often with much lower success rates [6, 7].

In the case of Tuberculosis, there are several potential competitive inhibitors of DHQS, sadly there are no conclusive result on most of them (no clear clinical results). However, recently it has come to light that DHQS is the main target of IMB-T130, and can potentially be used as a way to treat MDR-Tuberculosis [6].

In the case of *Acinetobacter baumannii* its ability to acquire multidrug resistance is far more vicious. It is able to acquire or upregulate resistance determinants representing it as one of the organisms that pose threats to present antibiotics. Recent computational studies have examined several potential candidate drugs targeting the Shikimate pathway, and have identified suitable drugs molecules from EMBL's bioactive molecules database (ChEMBL) [7].

Question 3

1) We load the **Escherichia coli core model** from BiGG Models and validate it, to check for errors:

a) i) Before validation:

The number of genes in the E.coli: 137

The number of reactions in the E.coli: 95

The number of metabolites in the E.coli: 72

=====

ii) After validation:

The number of genes in the E.coli: 137

The number of reactions in the E.coli: 95

The number of metabolites in the E.coli: 72

=====

The above output shows that the given model is reliable, and doesn't have any fatal errors!

b) The **cobra.io.sbml.validate_sbml_model** function performs the following checks [8]:

1. Checks whether SBML model can be accessed and read (checks for any missing model tags/annotations).
2. Checks some basic model properties such as reaction boundaries and compartment formulas.
3. Checks for schema errors (issues that violate file format rules)
4. Checks for all key fields of reactions, genes, etc. Checks for the consistency of units, and proper modelling practices as well.

5. Checks to see if all reaction bounds are feasible and no blocked reactions or dead-end metabolites exist.

c) To identify the gene knockout targets - (single genes, pairs and triplets) that would enhance the production of Ethanol, OptKnock was used.

i) Single Knockout

OptKnock result single knockout:

Processing solutions:  100%

	reactions	size	EX_eto_h_e	biomass	fva_min	fva_max
0	{CYTBD}	1	8.279455	0.211663	8.041861	9.061061
1	{EX_o2_e}	1	8.279455	0.211663	8.041861	9.061061
2	{O2t}	1	8.279455	0.211663	8.041861	9.061061

=====

ii) Double Knockout

OptKnock result double knockout:

Processing solutions:  100%

	reactions	size	EX_eto_h_e	biomass	fva_min	fva_max
0	{FORT, CYTBD}	2	16.586348	0.179985	15.956084	16.620484
1	{EX_o2_e, FORT}	2	16.586348	0.179985	15.956084	16.620484
2	{FORT, O2t}	2	16.586348	0.179985	15.956084	16.620484

=====

iii) Triple Knockout

OptKnock result triple knockout:

Processing solutions:  100%

	reactions	size	EX_eto_h_e	biomass	fva_min	fva_max
0	{ACKr, PYK, CYTBD}	3	18.612509	0.091291	17.977389	18.626383
1	{ACKr, EX_o2_e, PYK}	3	18.612509	0.091291	17.977389	18.626383
2	{ACKr, O2t, PYK}	3	18.612509	0.091291	17.977389	18.626383
3	{ACT2r, PYK, CYTBD}	3	18.612509	0.091291	17.977389	18.626383
4	{ACT2r, EX_o2_e, PYK}	3	18.612509	0.091291	17.977389	18.626383
5	{ACT2r, O2t, PYK}	3	18.612509	0.091291	17.977389	18.626383
6	{PTAr, PYK, CYTBD}	3	18.612509	0.091291	17.977389	18.626383
7	{PTAr, EX_o2_e, PYK}	3	18.612509	0.091291	17.977389	18.626383
8	{PTAr, O2t, PYK}	3	18.612509	0.091291	17.977389	18.626383
9	{PYK, EX_ac_e, CYTBD}	3	18.612509	0.091291	17.977389	18.626383
10	{EX_o2_e, EX_ac_e, PYK}	3	18.612509	0.091291	17.977389	18.626383
11	{O2t, EX_ac_e, PYK}	3	18.612509	0.091291	17.977389	18.626383

=====

2) Assumptions: Shown in Figure 3.2.1 are the exchange reactions of the **Escherichia coli core model**. For section 2.i and 2.ii, we are told to assume:

i) Glucose as the sole carbon source ($10 \text{ mmol} \cdot \text{gDW}^{-1} \cdot \text{h}^{-1}$)

ii) Acetate as the sole carbon source ($10 \text{ mmol} \cdot \text{gDW}^{-1} \cdot \text{h}^{-1}$)

To meet these requirements for 2.i, we set the “EX_glc__D_e” reaction’s upper and lower bounds to -10 (ensuring it is the sole carbon source).

And for 2.ii, we set “EX_glc__D_e” reaction’s lower bound to 0 (to ensure it is not a carbon source) and “EX_ac_e” reaction’s upper and lower bounds to -10 (ensuring it is the sole carbon source).

For 2.iii, we utilize minimal media.

Exchange Reactions of E.coli core model			
EX_ac_e	Acetate	0.0	1000.0
EX_acald_e	Acetaldehyde	0.0	1000.0
EX_akg_e	2-Oxoglutarate	0.0	1000.0
EX_co2_e	CO2 CO2	-1000.0	1000.0
EX_eto_h_e	Ethanol	0.0	1000.0
EX_for_e	Formate	0.0	1000.0
EX_fru_e	D-Fructose	0.0	1000.0
EX_fum_e	Fumarate	0.0	1000.0
EX_glc__D_e	D-Glucose	-10.0	1000.0
EX_gln__L_e	L-Glutamine	0.0	1000.0
EX_glu__L_e	L-Glutamate	0.0	1000.0
EX_h_e	H+	-1000.0	1000.0
EX_h2o_e	H2O H2O	-1000.0	1000.0
EX_lac__D_e	D-Lactate	0.0	1000.0
EX_mal__L_e	L-Malate	0.0	1000.0
EX_nh4_e	Ammonium	-1000.0	1000.0
EX_o2_e	O2 O2	-1000.0	1000.0
EX_pi_e	Phosphate	-1000.0	1000.0
EX_pyr_e	Pyruvate	0.0	1000.0
EX_succ_e	Succinate	0.0	1000.0

Figure 3.2.1: Exchange reactions in **Escherichia coli core model**

a) Given below is the pair of synthetic lethal genes in each case:

i) Glucose as the sole carbon source ($10 \text{ mmol} \cdot \text{gDW}^{-1} \cdot \text{h}^{-1}$)

{('b3919', 'b2284'), ('b4025', 's0001'), ('b3919', 'b2282'), ('b2417', 'b1817'), ('b0723', 'b3956'), ('b4025', 'b2280'), ('b0721', 'b3956'), ('b1761', 'b3212'), ('b3870', 'b1297'), ('b2987', 'b3493'), ('b3919', 'b2283'), ('b2417', 'b1818'), ('b0451', 's0001'), ('b4025', 'b2278'), ('b3919', 'b2276'), ('b4025', 'b2286'), ('b3956', 'b3236'), ('b0722', 'b3956'), ('b3919', 'b0767'), ('b3956', 'b4015'), ('b3919', 'b2029'), ('b3919', 'b2288'), ('b3919', 'b2285'), ('b3919', 'b1852'), ('b2417', 'b1819'), ('b3919', 'b2281'), ('b4025', 'b2287'), ('b4025', 'b2277'), ('b3919', 'b2279'), ('b3919', 's0001'), ('b4025', 'b2284'), ('b4025', 'b2282'), ('b3956', 's0001'),

('b3919', 'b2280'), ('b3919', 'b2278'), ('b3919', 'b2286'), ('b1761', 'b3213'), ('b0118', 'b1276'), ('b4025', 'b2283'), ('b4025', 'b2276'), ('b0724', 'b3956'), ('b0767', 'b4025'), ('b3919', 'b4025'), ('b4025', 'b2029'), ('b4090', 'b2914'), ('b4025', 'b2288'), ('b4025', 'b2285'), ('b3919', 'b2287'), ('b3919', 'b2277'), ('b4025', 'b1852'), ('b4025', 'b2281'), ('b2935', 'b2465'), ('b4025', 'b2279'). A total of 53 synthetic lethal gene pairs.

ii) Acetate as the sole carbon source ($10 \text{ mmol} \cdot \text{gDW}^{-1} \cdot \text{h}^{-1}$)

{('b1761', 'b3212'), ('b3870', 'b1297'), ('b2987', 'b3493'), ('b0979', 'b0733'), ('b3403', 'b0474'), ('b0979', 'b0734'), ('b3403', 'b1702'), ('b2976', 'b4014'), ('b0978', 'b0733'), ('b4232', 'b3925'), ('b2458', 'b2297'), ('b0118', 'b1276'), ('b1761', 'b3213'), ('b0978', 'b0734'), ('b3236', 'b0474'), ('b4090', 'b2914'), ('b3236', 'b1702'), ('b3236', 'b3956'), ('b2935', 'b2465')}. A total of 19 synthetic lethal gene pairs.

iii) Minimal media

{('b3919', 'b2284'), ('b4025', 's0001'), ('b3919', 'b2282'), ('b2417', 'b1817'), ('b0723', 'b3956'), ('b4025', 'b2280'), ('b0721', 'b3956'), ('b1761', 'b3212'), ('b3870', 'b1297'), ('b2987', 'b3493'), ('b3919', 'b2283'), ('b2417', 'b1818'), ('b0451', 's0001'), ('b4025', 'b2278'), ('b3919', 'b2276'), ('b4025', 'b2286'), ('b3956', 'b3236'), ('b0722', 'b3956'), ('b3919', 'b0767'), ('b3956', 'b4015'), ('b3919', 'b2029'), ('b3919', 'b2288'), ('b3919', 'b2285'), ('b3919', 'b1852'), ('b2417', 'b1819'), ('b3919', 'b2281'), ('b4025', 'b2287'), ('b4025', 'b2277'), ('b3919', 'b2279'), ('b3919', 's0001'), ('b4025', 'b2284'), ('b4025', 'b2282'), ('b3956', 's0001'), ('b3919', 'b2280'), ('b3919', 'b2278'), ('b3919', 'b2286'), ('b1761', 'b3213'), ('b0118', 'b1276'), ('b4025', 'b2283'), ('b4025', 'b2276'), ('b0724', 'b3956'), ('b0767', 'b4025'), ('b3919', 'b4025'), ('b4025', 'b2029'), ('b4090', 'b2914'), ('b4025', 'b2288'), ('b4025', 'b2285'), ('b3919', 'b2287'), ('b3919', 'b2277'), ('b4025', 'b1852'), ('b4025', 'b2281'), ('b2935', 'b2465'), ('b4025', 'b2279')}. A total of 53 synthetic lethal gene pairs.

b) The table below shows which pairs of synthetic lethal genes fall under which category (considering the first 5 genes in each case):

S.No	Synthetic Lethal Genes	Common to all cases	Case (i)	Case (ii)	Case (iii)
1	[b1761, b3213]	Yes	Yes	Yes	Yes
2	[b1761, b3212]	Yes	Yes	Yes	Yes
3	[b3956, b4015]	No	Yes	No	Yes
4	[b0118, b1276]	No	Yes	No	Yes
5	[b2417, b1819]	No	Yes	No	Yes
6	[b3236, b0474]	No	No	Yes	No
7	[b3236, b3956]	No	No	Yes	No
8	[b3236, b1702]	No	No	Yes	No

{('b4090', 'b2914'), ('b1761', 'b3212'), ('b3870', 'b1297'), ('b2987', 'b3493'), ('b1761', 'b3213'), ('b2935', 'b2465'), ('b0118', 'b1276')}, are the 7 common synthetic lethal genes in all three cases.

c) The above table and the fact that only 7 synthetic lethal genes are common in all the three cases indicates that synthetic lethality is dependant on the media. Most organisms have evolved to utilize Glucose as their primary carbon source (due to its typical abundance in nature), as such when it is the only carbon source in the media, KO of 2 redundant but critical genes in Glucose Metabolism Pathway will lead to the death of the cell, and since Glucose is the cell's primary carbon source, there are likely more targets in this case.

However, in the case of Acetate as the sole carbon source, since it is only during times of crisis that a cell usually suffers such hardships, as such it appears to be more robust (that is, there are far fewer synthetic lethal gene pairs in this case 19 as opposed to 53 in the other cases). The 7 common synthetic lethal genes are most likely related to pathways that the cell cannot survive without (Cell wall maintenance, Essential amino acid synthesis, DNA repair, etc.)

3) a) We choose to analyse the synthetic lethal pair of genes b1761 and b3213.

b1761 - **NADP-specific Glutamate dehydrogenase** [9] ; b3213 - **Glutamate synthase [NADPH]** [10]

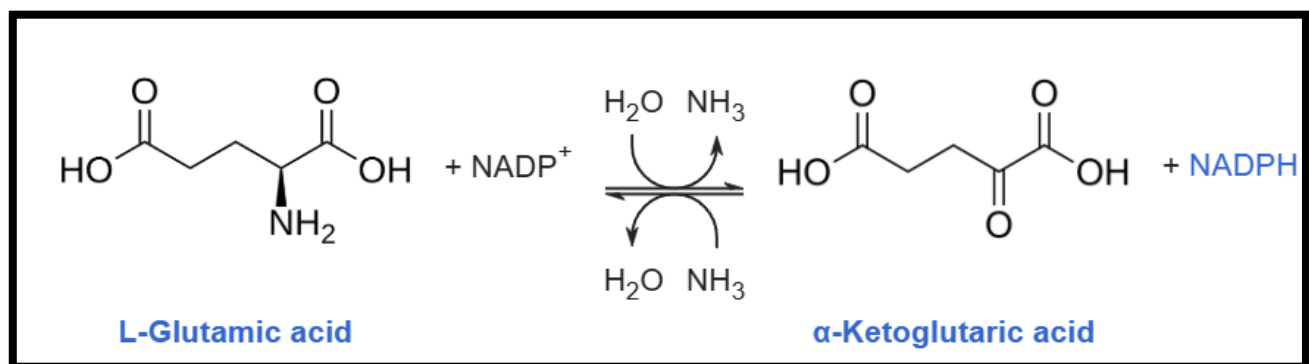


Figure 3.3.a.1: Glutamate dehydrogenase reaction

The GDH pathway, is a glutamate synthesis pathway and is active when the organism is stressed for energy (glucose-limited conditions), because the enzyme does not use ATP. The affinity of Glutamate dehydrogenase to ammonia is also very low, which implies that in most normal conditions, the forward reaction occurs to produce α -ketoglutarate. However, during high stress conditions or excess NH_3 in the environment, this becomes the primary pathway in the production of glutamate [11,13].

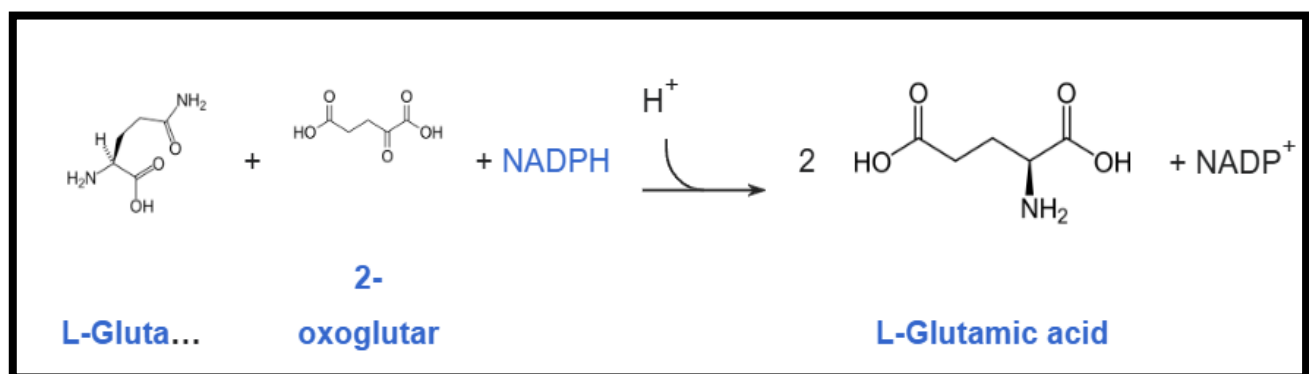


Figure 3.3.a.2: Glutamate synthase [NADPH] reaction

The Glutamate synthase pathway (also known as Glutamine oxoglutarate aminotransferase or GOGAT for short) manufactures glutamate from glutamine and α -ketoglutarate, and thus along with glutamine synthetase (abbreviated GS) [12]. As shown in the above reaction in Figure 3.3.a.2, this enzyme converts the glutamine produced by the GS enzyme into glutamate. The GS enzyme utilizes 1 ATP molecule to convert glutamate into an acyl-phosphate intermediate as shown below in Figure 3.3.a.3. This ultimately produced the required glutamine

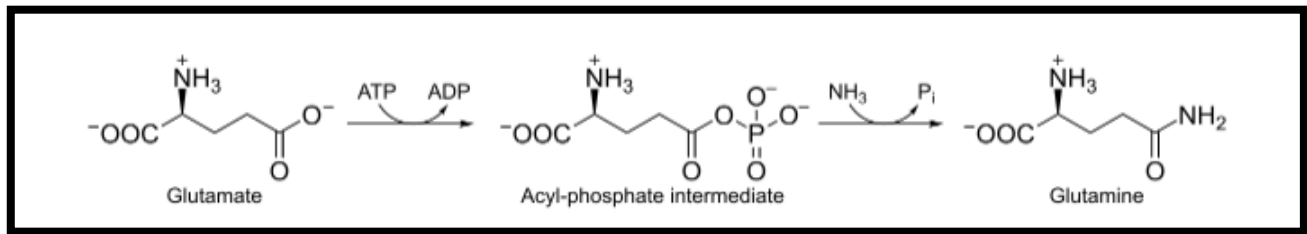


Figure 3.3.a.3: Glutamine synthetase (GS) reaction

Since, GDH has a low K_m and glutamate production isn't favoured in that pathway, this serves as the primary pathway for glutamate production.

The simultaneously deletion of both genes leads to the cell becoming incapable of producing glutamate (which is a key amino acid). Thus, this would lead to cell death (unless the media supplies glutamate).

b) However, the existence of this redundant pathway allows for the mutants lacking genes of either pathway still survive, but mutants lacking both require external glutamate for growth. Thus, one or the other pathway is necessary for glutamate synthesis, but not both [13].

The redundant route here would be the GDH pathway (since it functions only under cases of extreme stress), and this back-up/redundant route gets removed along with the removal of the main GOGAT pathway, effectively preventing cell from producing any glutamate.

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Discussions:

1. Regarding fast-sl, I was having issues with the !sed command since I could not use it locally on my Windows machine. I had asked Sahaj how to run the fast-sl patch locally, upon which he helped me in reading and rewriting the specific section of the file.
 2. Discussed with Skanda and Krushanu on what is an apt threshold to quantify an essential gene for Question 2.
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